



Smart Scan Pro Light

AI-Powered Multispectral Forensic Imaging System

MSc Forensic Science Project & Research Funding Presentation

Smart Scan Pro Light – Overview



A portable AI-powered forensic imaging device designed to improve crime scene investigation by combining multispectral illumination, high-resolution imaging, artificial intelligence, and automated forensic documentation into a single handheld instrument.

The device integrates multiple light wavelengths (UV, visible, and infrared), an embedded camera, an AI processing unit, and automated report generation software — enabling investigators to visualize hidden evidence and generate standardized forensic reports directly at the crime scene.



16 MP AI Camera
Integrated high-resolution imaging



Multispectral LEDs
UV, visible & infrared illumination



Real-Time AI
On-device image enhancement



Auto PDF Reports
One-click forensic documentation



Battery Powered
Rechargeable

SMART SCAN PRO LIGHT

AI POWERED FORENSIC IMAGING & REPORT GENERATOR

SEE BEYOND THE VISIBLE. DETECT. ANALYZE. DOCUMENT.

MULTISPECTRAL LED ARRAY

- UV 365 nm
- UV 302 nm
- BLUE 450 nm
- GREEN 530 nm
- AMBER 590 nm
- RED 630 nm
- WHITE
- IR 850 nm

16 MP CAMERA MODULE

- High Resolution
- Auto Focus
- Macro Lens
- Polarizing Filter

ERGONOMIC DESIGN

- Comfort Grip
- Lightweight
- Rugged Body

POWER SYSTEM

- 12V 5000 mAh Battery
- USB-C Charging
- 4-5 Hours Working

ON DEVICE TOUCH DISPLAY

- Live View
- Mode Selection
- Brightness Control
- Capture
- Settings

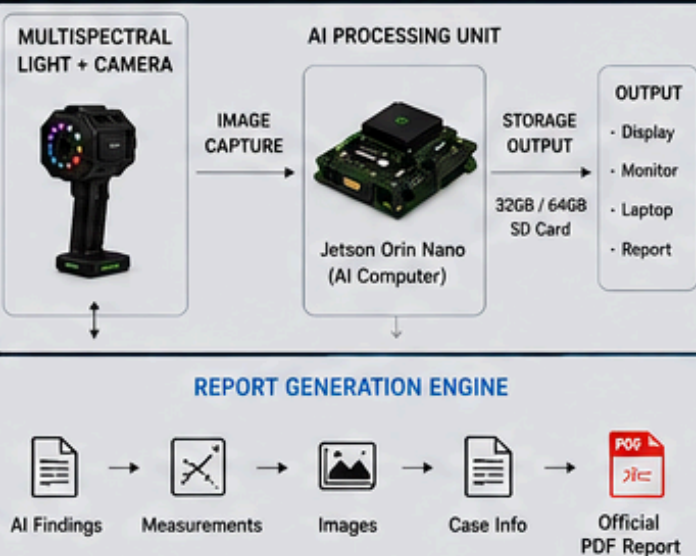
WHAT IT DOES

- Multispectral Imaging
- AI Based Evidence Detection
- Image Enhancement
- Measurement Tools
- Report Generation
- Connect to Monitor / Laptop
- Wi-Fi & USB Connectivity

LIVE VIEW ON MONITOR / LAPTOP



SYSTEM ARCHITECTURE



AI WORKFLOW



SMART SCAN PRO LIGHT FORENSIC IMAGING REPORT

CASE INFORMATION

Case No. : 2024/OS/00123
Date & Time : 20 May 2024 14:35:22
Location : Delhi, India
Examiner : Inspector R. Sharma
Device ID : SSL-AI-2024-0007

IMAGING PARAMETERS

Mode : UV 365 nm
Resolution : 16 MP
Exposure : 1/125 sec
ISO : 200
Temperature : 28.4 °C
Humidity : 48 %

AI FINDINGS

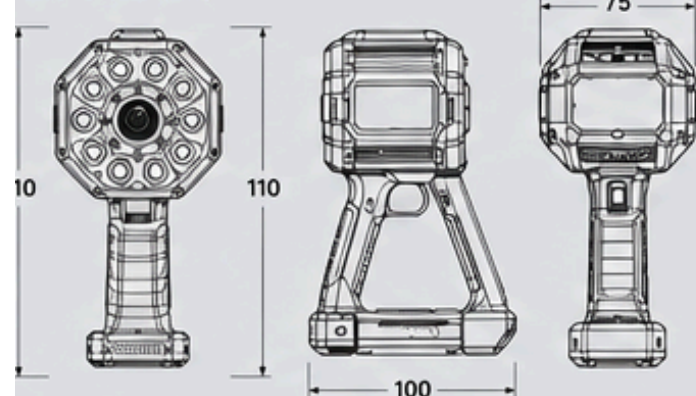
Evidence	Original	Enhanced	Annotated
Blood Stain			
Fingerprint			
Fibre			
Hair			
Shoeprint			
Document Alteration			

AI INTERPRETATION

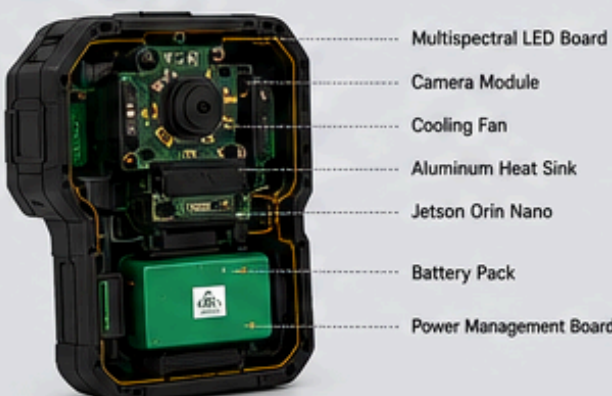
The stain exhibits fluorescence characteristics consistent with a possible biological fluid under 365 nm excitation. Confirmation by laboratory testing is recommended.



DIMENSIONS (mm)



INTERNAL DESIGN



CONNECTIVITY

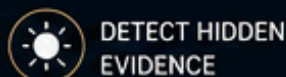
- HDMI (To Monitor)
- USB-C (To Laptop / Charging)
- Wi-Fi (Wireless Transfer)
- SD Card (Storage)

IN THE BOX



TECHNICAL SPECIFICATIONS

- Camera : 16 MP, 1080p Video
- LEDs : 8 Multispectral Modes
- Wavelengths : UV, Blue, Green, Amber, Red, White, IR 850 nm
- Display : 3.5" Touchscreen
- Storage : 32GB / 64GB SD Card
- Connectivity : HDMI, USB-C, Wi-Fi
- Battery : 12V 5000 mAh Li-ion
- Working Time : 4 - 5 Hours
- Weight : ~600 g



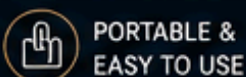
DETECT HIDDEN EVIDENCE



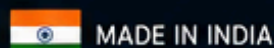
AI POWERED ANALYSIS



INSTANT REPORT GENERATION

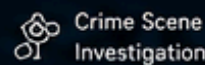


PORTABLE & EASY TO USE



MADE IN INDIA

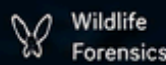
IDEAL FOR



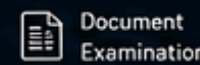
Crime Scene Investigation



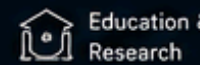
Forensic Laboratories



Wildlife Forensics



Document Examination

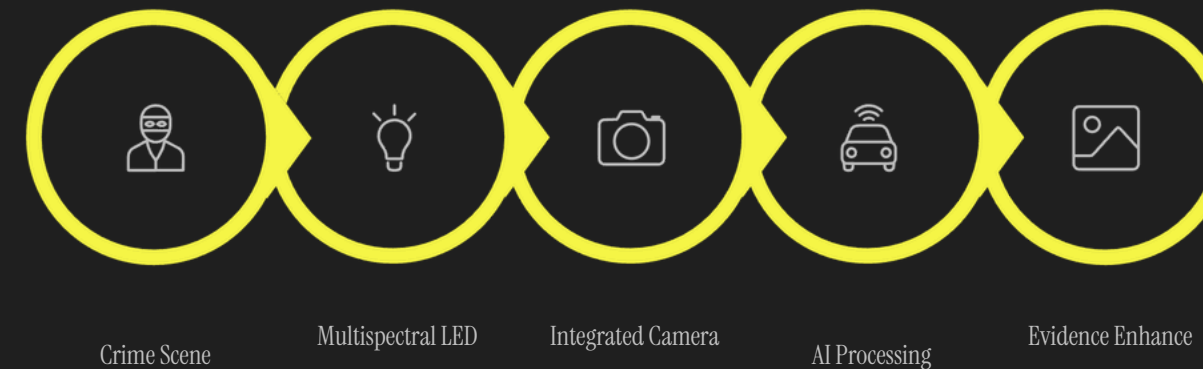


Education & Research

System Design and Working Mechanism



Complete Workflow



The end-to-end pipeline from raw scene capture to court-ready documentation, all processed on-device in real time.

Hardware Components

Multispectral LED Array

UV, Blue, Green, Red, White, IR wavelengths

High-Resolution Camera

16 MP integrated imaging sensor

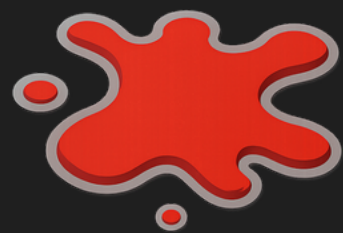
Embedded AI Processor

Raspberry Pi 5 or NVIDIA Jetson Orin Nano

Connectivity

HDMI, USB-C, Wi-Fi, touchscreen interface

Applications of Smart Scan Pro Light



Biological Evidence

Blood and biological stain visualization, latent fingerprint imaging, fibre and hair examination



Physical Marks

Footwear and tool mark documentation, document examination, laboratory evidence photography



Specialist Investigations

Wildlife forensic investigations, educational and forensic research, preliminary field documentation

Key Advantages



All-in-One Device

Combines multiple forensic tools into one portable instrument



Faster Documentation

AI-assisted enhancement reduces investigation time significantly



Cost-Effective

Budget-friendly prototype with user-friendly interface



Budget

Raspbary pi = 9000

ssd card = 1200

Power supply = 700

Display = 7000

3d print = 2000

Sunboard = 500

Lights = 2000

Tools = 5000

Camera = 4000

App = 16000

AI = 5000

Battery = 2000

Wires = 400

Buttons = 800

Switches = 600

Cables = 1500

Batttery charger =1000

Power adapter with cable and bms = 3500

Sd card reader = 1500

Addasives = 500

Micelleneous= 2000

Total= 66200 Only

Technical Specifications & Innovative Features

Technical Specifications

Component	Specification
Camera	16 MP integrated sensor
Display	3.5-inch touchscreen
Light Sources	UV 365 nm, Blue 450 nm, Green 530 nm, Red 630 nm, White, IR 850 nm
AI Platform	Raspberry Pi 5 or NVIDIA Jetson Orin Nano
Connectivity	HDMI, USB-C, Wi-Fi
Battery	Rechargeable Li-ion, 4-5 hours
Weight	Approximately 600-800 g

Innovative Features

- ➔ AI-Based Evidence Enhancement
Real-time processing with OpenCV, YOLO & TensorFlow
- ➔ Automatic Measurement Tools
On-screen scale and spatial reference overlays
- ➔ One-Click Report Generation
Standardized PDF forensic reports at the scene
- ➔ Modular & Open-Source
Upgradeable hardware and open-source software architecture
- ➔ Live Monitor Connectivity
HDMI output for real-time display on external screens

Future Scope and Conclusion

Future Development Roadmap

- 1 — Near-Term Upgrades
Thermal imaging integration, hyperspectral imaging, 3D evidence scanning, GPS-enabled evidence logging
- 2 — Platform Expansion
Cloud-based forensic database, mobile application integration, voice-controlled operation
- 3 — Advanced AI & Legal
AI-powered crime scene reconstruction, advanced ML models for evidence classification, court-ready documentation support

Conclusion

Smart Scan Pro Light demonstrates how artificial intelligence, multispectral imaging, and automated documentation can be combined into a portable forensic imaging platform. As a low-cost proof-of-concept, it has the potential to improve crime scene documentation, support forensic education and research, and serve as a foundation for future development into a validated commercial forensic imaging system.

Smart Scan Pro Light – **See Beyond the Visible. Detect. Analyze. Document.**

